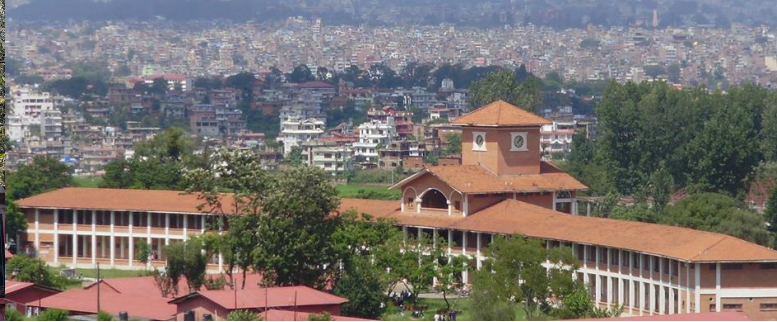

Replication in Economics Research Basics

Replication 101

Practical notes on replication packages, reproducibility

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Replication in Economics Research Basics

Focus: why replication packages matter

and how to build a package that another researcher can actually run.

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Why replication packages matter

- A replication package is the bundle of data, code, documentation, and computing instructions that allows another researcher to reproduce the published results.
- In economics, replication matters for transparency, verification, and reuse.
- A well-prepared package reduces ambiguity about data construction, estimation choices, and the path from raw inputs to final tables and figures.
- It also turns a paper into something that can be checked, learned from, and extended by others.

Source: Hamermesh (2007); AEA Data Editor reports (2019, 2024).

The key principle: computational empathy

Core idea

Write the package for a careful graduate student working on a clean machine, not for the author who already knows the project history.

Ask yourself

Could someone else run this without my laptop, local folders, or undocumented manual steps?

Do:

- provide one master run file
- pin software versions where important
- explain restricted-data branches honestly

Avoid:

- manual clicking and hand edits
- hidden preprocessing
- missing dependencies
- output folders that do not reproduce cleanly

Source: AEA Data Editor practical guidance on preparing files for verification.

A practical structure for a replication package

Folder / file	What it should do
`README`	Explain software, runtime, execution order, data availability, and table/figure mapping
`data/raw/`	Store original data or document how restricted/proprietary data are accessed
`data/analysis/`	Hold cleaned or constructed datasets used in estimation
`code/`	Include scripts for cleaning, estimation, diagnostic tests, and figures
`run_all.do` / `main.R`	Provide a single entry point for rerunning core results
`results/`	Save generated tables, figures, logs, and diagnostics

Rule of thumb

If a stranger cannot tell what to run, in what order, with which software, and what successful output should look like, the package is not ready.

Common failure points

- Late assembly: authors organize files only after acceptance.
- Weak README: no runtime, software, or output map.
- Confidential-data confusion: missing explanation of what can and cannot be shared.
- Code rot: scripts accumulated across multiple machines no longer run from scratch.
- Hidden workflow steps: manual preprocessing exists outside the documented pipeline.

Editor-side implication

A package may seem complete to the author and still fail verification because the files do not reproduce the claimed results on an external machine.

Replication 101 workflow for students

1. Choose a paper with feasible data access and manageable software requirements.
2. Map the pipeline from raw data to cleaned data, estimates, and final exhibits.
3. Reproduce the headline tables and figures before attempting extensions.
4. Record every deviation from the published workflow.
5. Re-run the full pipeline from a clean session.
6. Write a README for another person, not for your future self.
7. Archive outputs and logs so failures can be diagnosed.

Pedagogical value

Replication teaches empirical design by forcing the researcher to understand data construction, coding choices, and the logic linking methods to results.

Takeaway checklist

- Can someone rerun the package without access to my personal machine?
- Is there one obvious command or script that reproduces the core results?
- Does the README explain data rights, software, runtime, and output mapping?
- Are restricted or proprietary components documented clearly and honestly?
- Did I rerun everything from scratch after cleaning folders and outputs?

A good replication package is not extra paperwork;
it is part of credible empirical research.

Selected sources

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